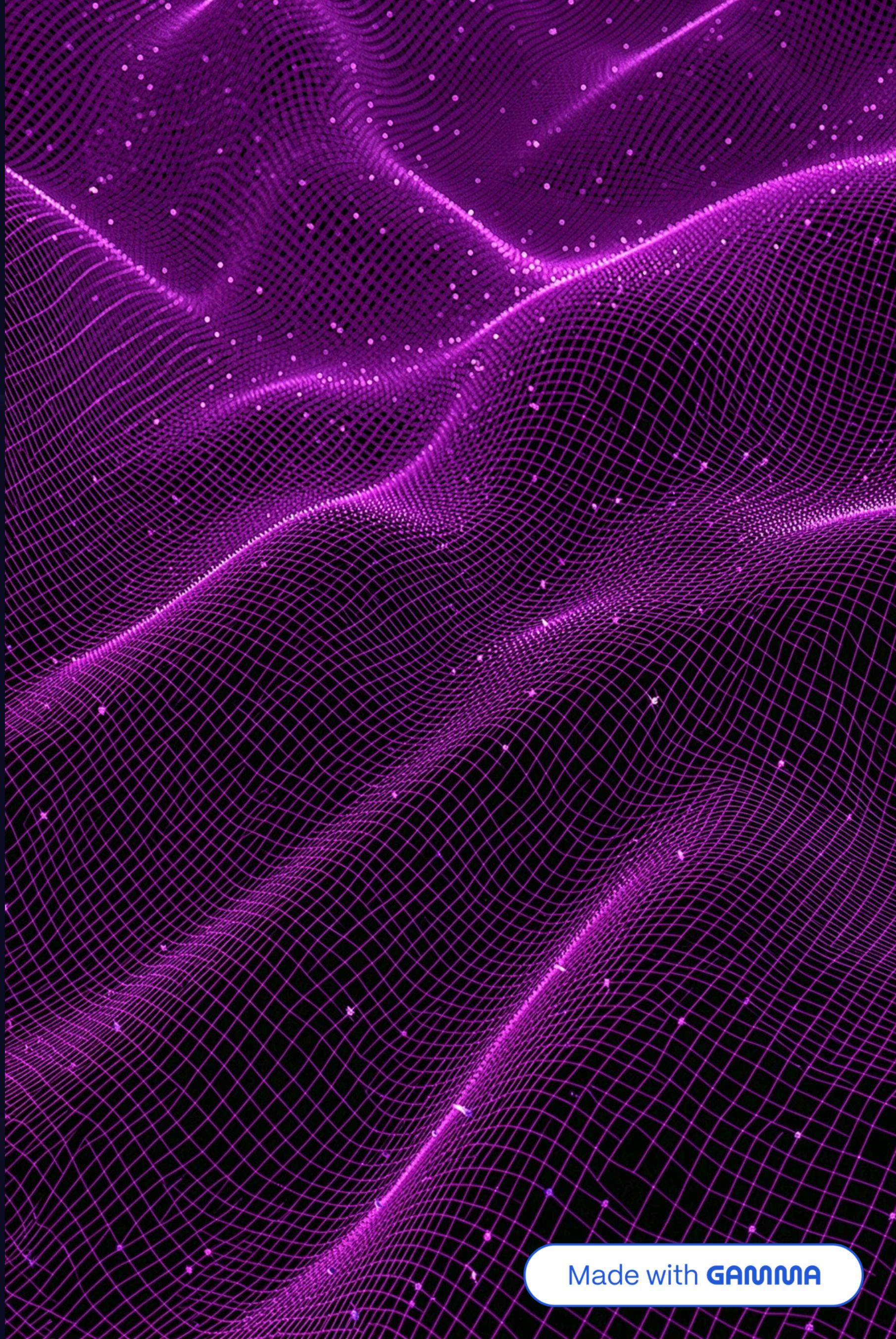
App User Behavior Segmentation

Unlocking deeper insights with Unsupervised Machine Learning — grouping 50,000 users into meaningful segments without predefined labels.

FINAL PROJECT REPORT

K-MEANS CLUSTERING

USER ANALYTICS



The Problem: One-Size-Fits-All Marketing Fails

Generic campaigns miss behavioral nuances, and arbitrary thresholds create artificial segments that ignore real user journeys.

Missed Nuances

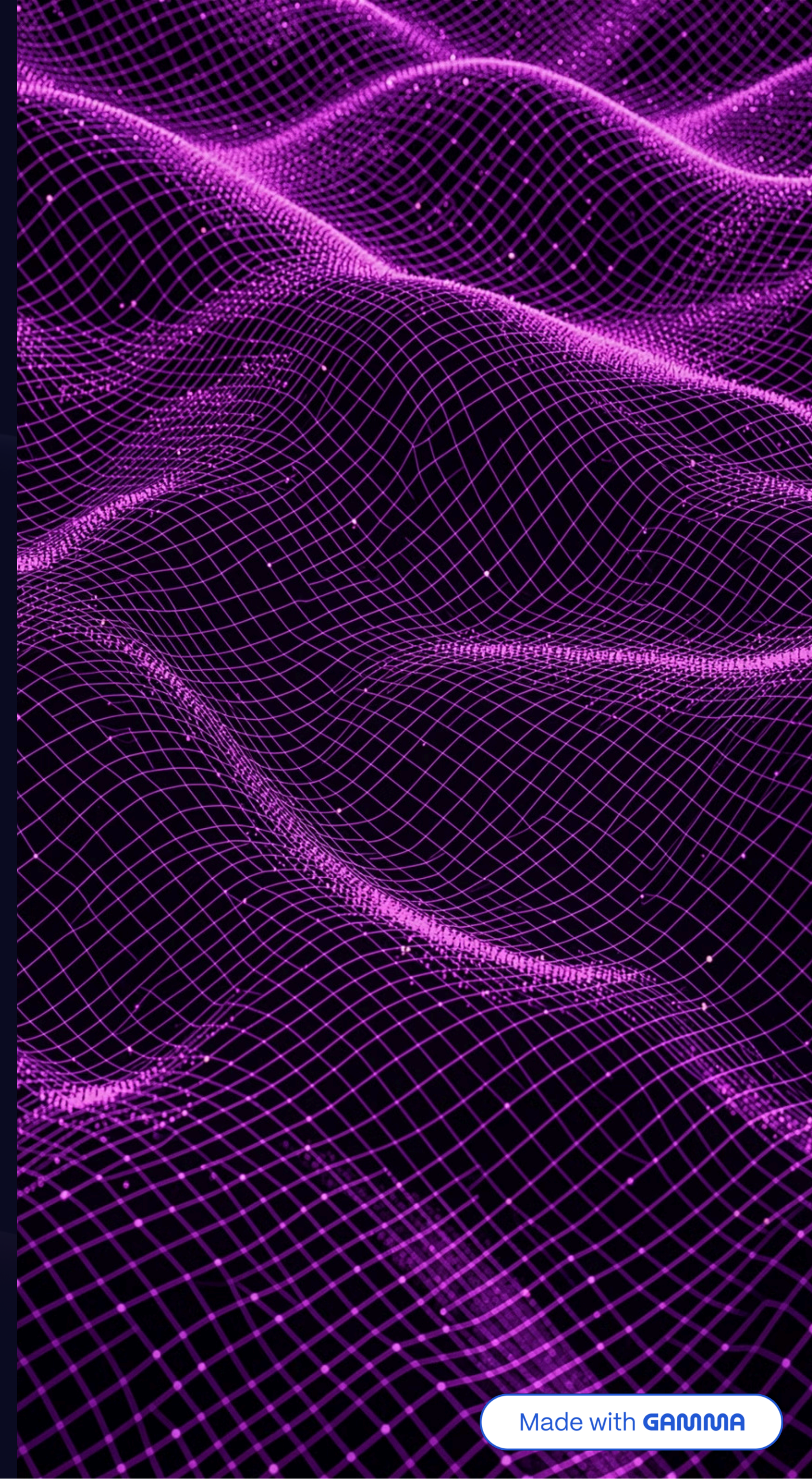
Generic campaigns treat all users identically, ignoring behavioral diversity.

Artificial Thresholds

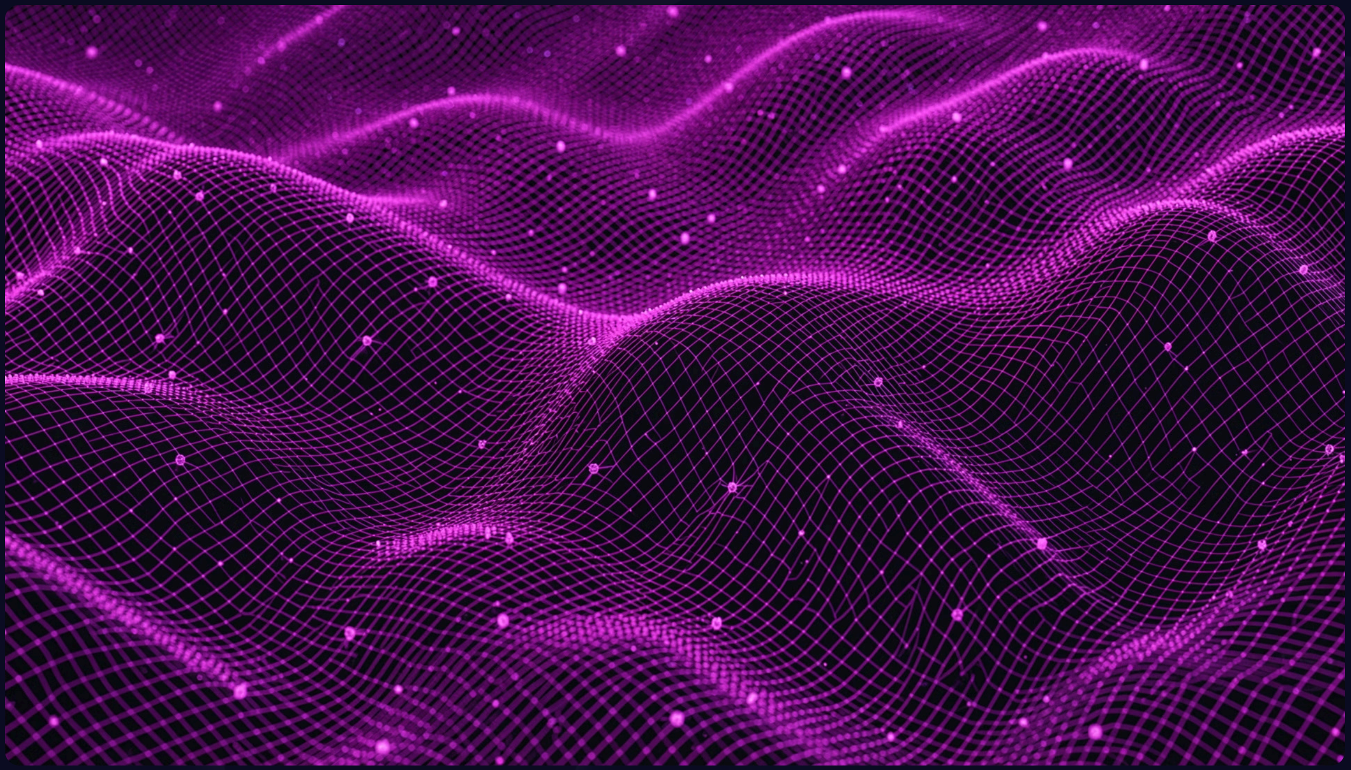
Rules like "5+ sessions" create rigid segments that don't reflect real journeys.

Wasted Spend

Poor personalization drives low conversion and inefficient marketing investment.



The Solution: Unsupervised Learning for Natural Groupings



Unsupervised learning algorithms discover hidden patterns in unlabeled data — no manual rules required.

No Predefined Labels

The algorithm learns structure directly from raw behavioral data.

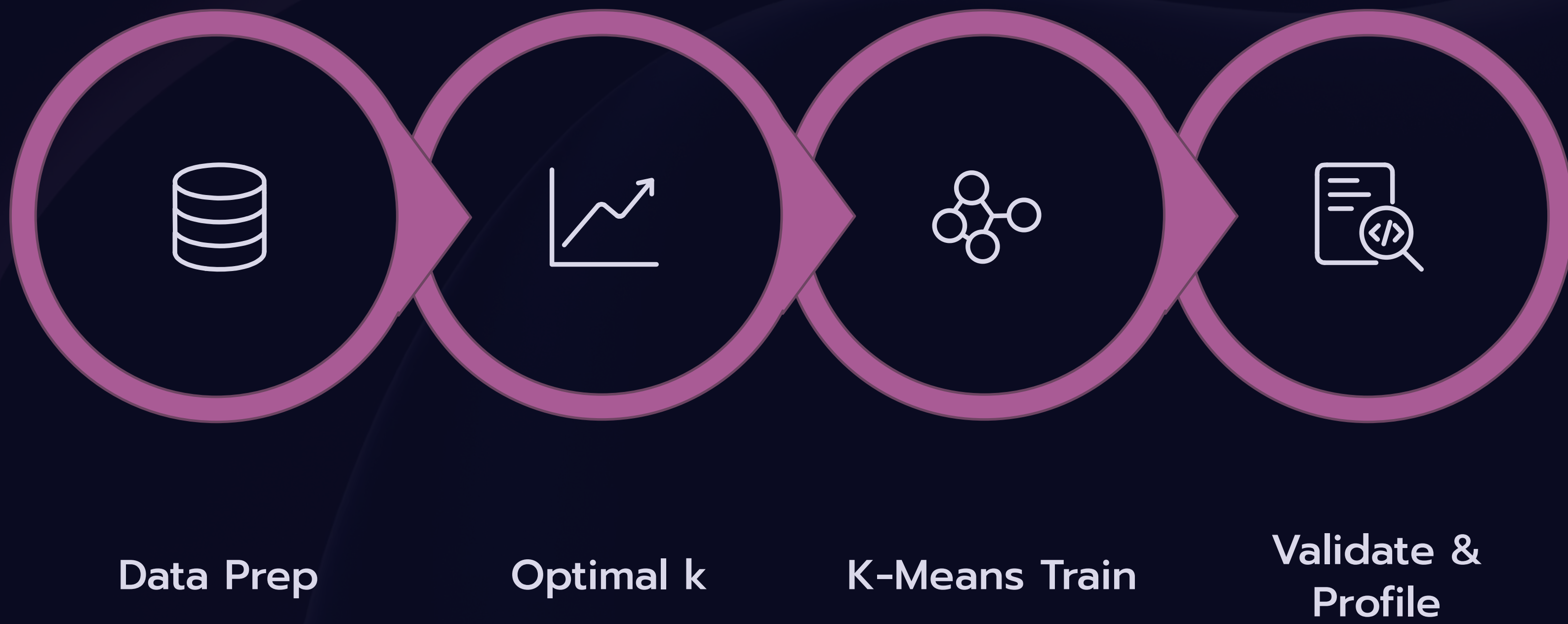
K-Means Clustering

Groups users by inherent behavioral similarities, not arbitrary rules.

Truly Distinct Segments

Reveals natural user groups that manual segmentation would miss.

Methodology: From Raw Data to Actionable Segments



The 13-step pipeline processes 50,000 user records — from raw data through cleaning, scaling, clustering, and validation — to produce actionable behavioral segments.

Unveiling the 4 User Segments

K-Means clustering revealed four distinct behavioral groups, each with unique engagement patterns.

Cluster 0 — High Engagement

High session frequency, long duration, low churn risk.

Cluster 1 — Moderate Engagement

Balanced, consistent usage with moderate engagement.

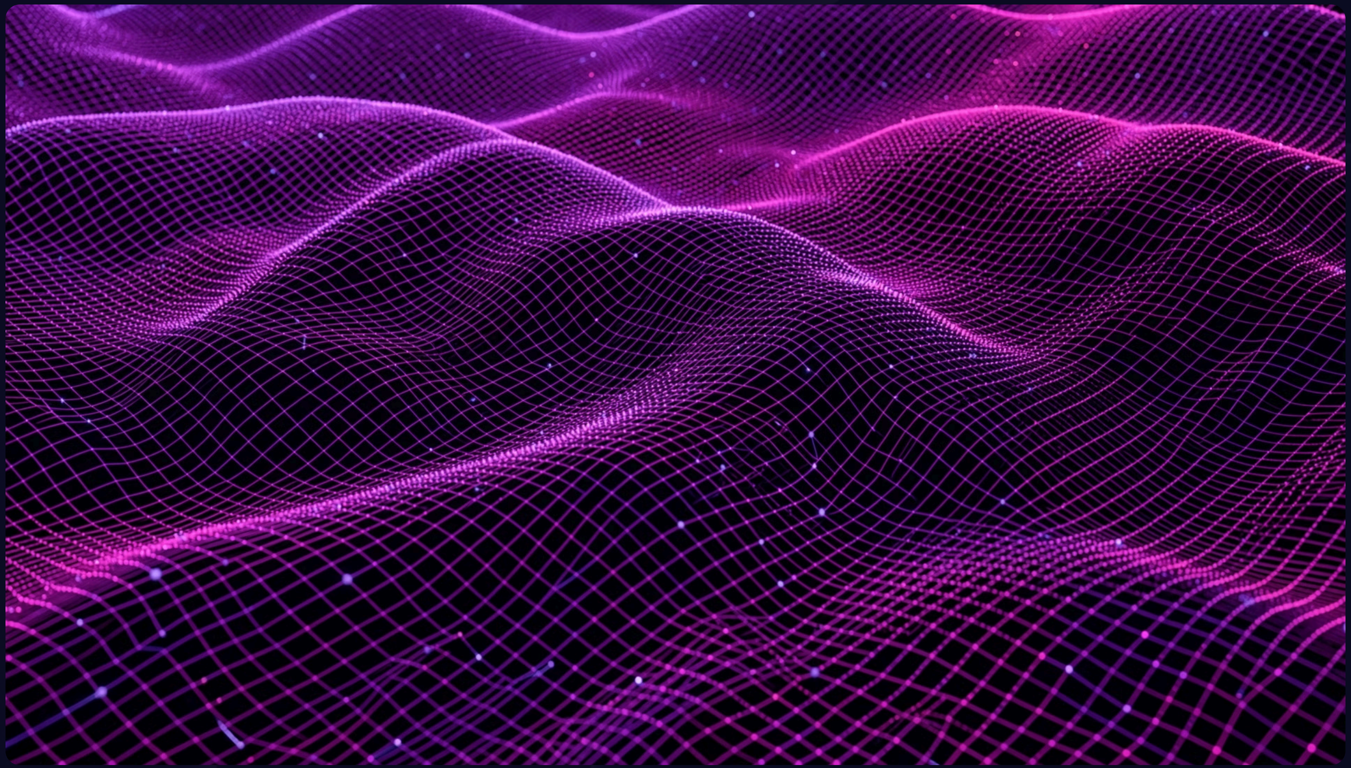
Cluster 2 — At-Risk Users

Low activity, high churn risk — needs immediate intervention.

Cluster 3 — Occasional Users

Irregular usage and low interaction frequency.

Visualizing Cluster Separation with PCA



Dimensionality Reduction

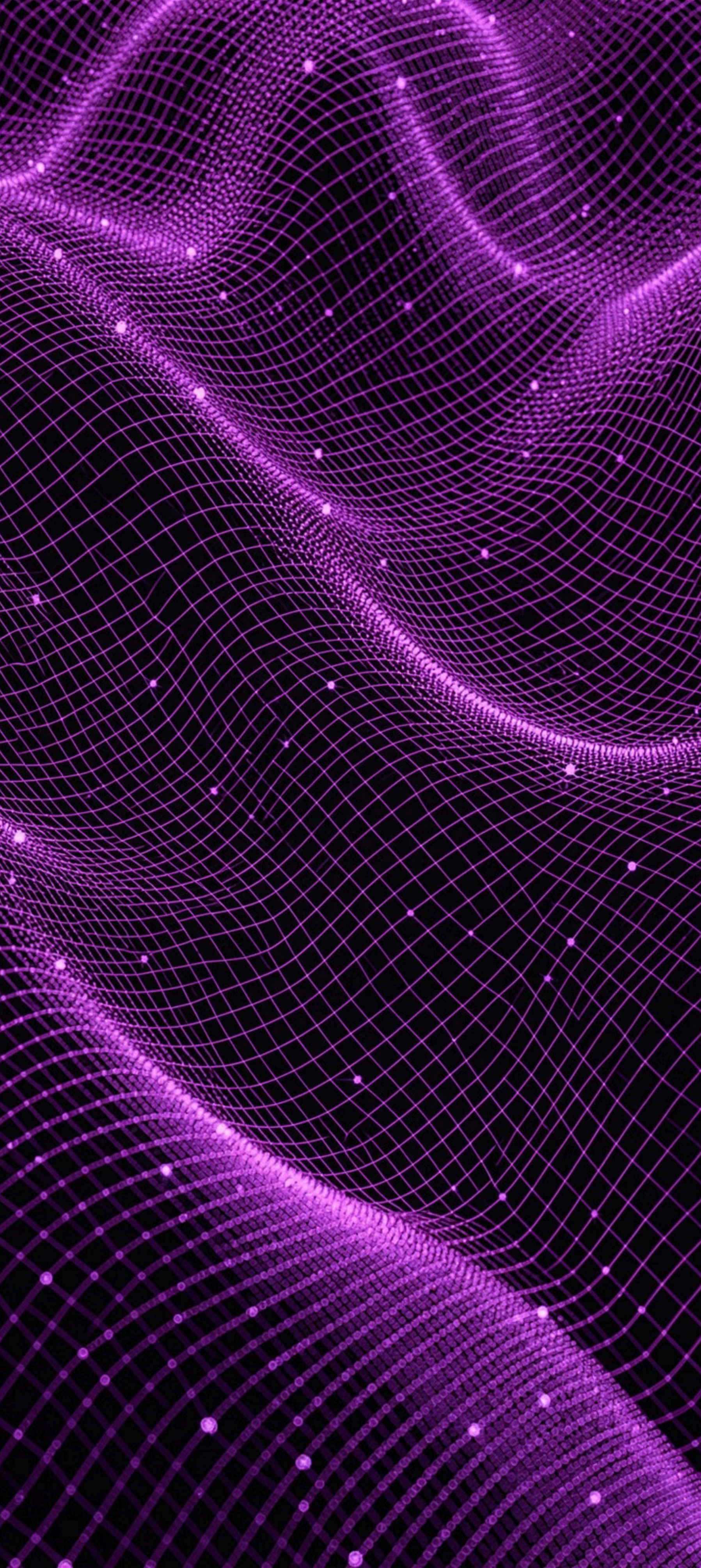
Principal Component Analysis reduced 9 behavioral features to 2 dimensions for clear visualization while preserving meaningful variance.

Clear Separation

PCA plot shows distinct boundaries between all four clusters.

Validated Cohesion

Silhouette Score confirmed strong intra-cluster similarity and inter-cluster separation.



Business Impact: Smarter Strategies, Better Results

Personalized Campaigns

Tailor messages and offers to each segment's specific behavior and needs.

Improved Retention

Proactively engage at-risk users with targeted re-engagement strategies.

Increased Conversion

Deliver relevant promotions to transactional users and guide newcomers.

Optimized Resources

Focus marketing spend on segments with the highest potential value.

Case Study & Deployment Readiness

Case Study Highlight

K-Means segmentation on app data identified a previously unknown "Loyal Discount Seekers" group. Targeted promotions for this segment drove a **15% increase in repeat purchases** within one quarter.

Production-Ready Pipeline

01

Model Serialized

K-Means model and StandardScaler saved via pickle for future predictions.

02

New User Assignment

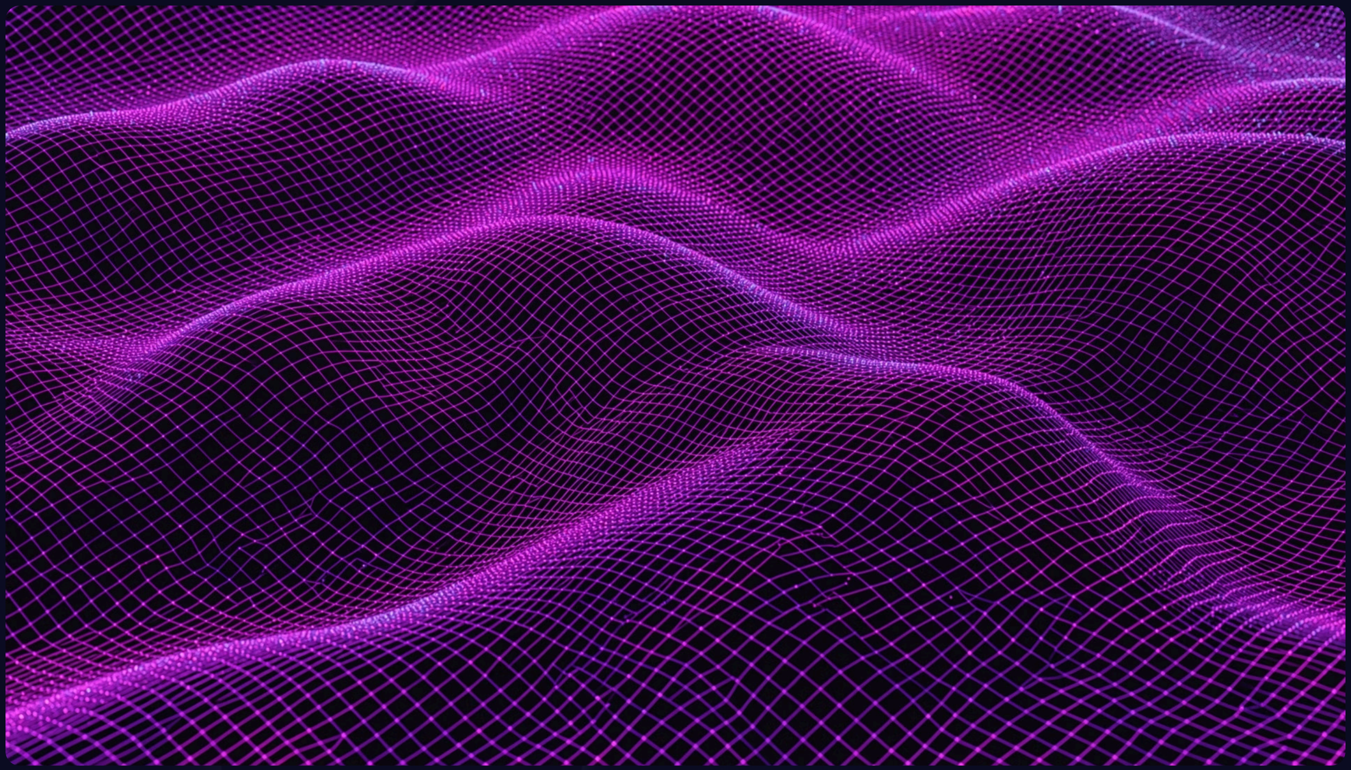
Incoming users assigned to clusters in real time using the trained model.

03

Dashboard Integration

Results exportable to Power BI or Tableau for stakeholder reporting.

Tech Stack & Tools



Core Technologies

Python

Core programming language for data processing and modeling.

Pandas & NumPy

Data manipulation, cleaning, and numerical computation.

Scikit-learn

K-Means, StandardScaler, Silhouette Score, and PCA implementation.

Matplotlib & Seaborn

Elbow plots, heatmaps, and PCA scatter visualizations.

The Future: Continuous, Evolving Segmentation

User behavior evolves — segmentation must too. This project lays the foundation for a truly data-driven approach to engagement and growth.

Advanced Algorithms

Explore DBSCAN and Hierarchical Clustering for nuanced segment discovery.

Real-Time Streaming

Integrate live data pipelines for dynamic, up-to-the-minute segmentation.

Deep Learning

Apply autoencoders and neural networks for richer behavioral prediction.

Dashboard Integration

Connect outputs to Power BI or Tableau for automated stakeholder reporting.